Payload-Aware Two-Tier UAV Route Planning for Last-Mile Delivery

Thirumurugan RA (3122235001149), Shrika SP Thota (3122235001125), Meghna Manimaran (3122235001079)

Guide Name : Sujaudeen N, Assistant Professor, Department of Computer Science and Engineering

Problem Domain

Autonomous UAV systems for medical and emergency last-mile delivery, where route planning must simultaneously satisfy conflicting objectives (travel distance, battery consumption, and flight risk) under dynamic field conditions without on-site expert supervision.

Motivation:

- Fragile payloads require low-vibration, low-risk paths; emergency medicine requires minimum-time routes; disaster-relief supplies require energy-efficient coverage. Every payload type demands a fundamentally different route trade-off.
- No current planner adjusts its routing priorities to match what is being carried. The same fixed objective weights are applied to every mission, forcing field operators to manually re-calibrate between every deployment, which is impractical and error-prone in time-critical emergency scenarios.

Problem Statement

Existing UAV route planners encode distance, energy, and flight risk into **a single hand-tuned weighted-sum scalar** before path search begins, permanently fixing the objective trade-off regardless of what is being carried. When payload context changes from fragile diagnostic samples requiring vibration-minimal routing to time-critical emergency medicine requiring shortest-path routing an expert must manually re-tune planner weights and re-run the search from scratch, an intervention that is **infeasible in field conditions.**

Our proposed system resolves this by separating Pareto front generation from mission-context-driven route selection; payload metadata (fragility class, temperature sensitivity, delivery urgency) has never been used as the sole input to autonomous Pareto point selection in the UAV delivery literature, leaving a structural gap this project directly addresses.

Existing Work

Chokkalingam & Ganesan (2026) - *RAIRO-Oper. Res.*: https://doi.org/10.1051/ro/2025159

- **Technique:** Multi-objective UAV path planning using adaptive constraint handling with entropy-based criteria to select solutions from the Pareto front.
- **Limitation:** Entropy criterion selects solutions based on front diversity alone; payload metadata (fragility, urgency, thermal sensitivity) is never incorporated into the selection process.

Datsko et al. (2024) - *IEEE RA-L*: https://ieeexplore.ieee.org/document/10414185

- **Technique:** Diversity-driven Pareto selection for UAV routing, optimising geometric spread of candidate solutions across the multi-objective space.
- **Limitation:** Selection is guided entirely by solution diversity; delivery context such as payload type, time-window, or priority class has no influence on which route is chosen.

Ntakolia et al. (2023) - *J. Intell. Robot. Syst.*: https://doi.org/10.1007/s10846-023-01945-9

- **Technique:** Crowding-distance-based Pareto point selection within a multi-objective UAV path planner to maximise solution spread across objective space.
- **Limitation:** Crowding distance cannot encode mission-specific requirements; payload fragility, temperature sensitivity, or urgency class cannot influence route selection at any stage.

Proposed Work

Project Objectives

- **Tier 1 - Deterministic Pareto Front Construction:** Implement a multi-objective A*/Dijkstra graph-search that pre-computes and stores a well-distributed Pareto-optimal set of routes across route distance, energy consumption, and flight risk for a defined delivery region; the front is reused across successive missions without re-running the planner.
- **Tier 2 - Payload-Driven Pareto Selection:** Implement a structured payload metadata interpreter that reads four operator-assigned tags (fragility class, temperature sensitivity, delivery time-window, priority level) and maps them to a single Pareto-front point autonomously no weight re-tuning at any stage.
- **Ground-Station Dashboard:** Build an operator interface where a payload tag is entered once and a ready-to-fly mission file (compatible with ArduPilot Mission Planner) is generated without further manual input.
- **Hardware Validation:** Demonstrate the end-to-end pipeline on a quadcopter UAV through real outdoor flight tests, measuring route adherence and end-to-end latency from payload tagging to mission execution.

Research Objectives

- Formulate multi-objective UAV route planning as a structured **Pareto front pre-computation** problem over a delivery airspace graph, theoretically distinguishing the deterministic graph-search approach from stochastic population-based alternatives
- Design a **payload metadata schema** and a principled rule-based mapping that translates mission context (fragility, urgency, thermal constraints) into Pareto point selection without any expert weight tuning.
- **Benchmark Tier 1** route quality and Pareto front coverage against single-objective A* and MOPSO baselines on identical delivery scenarios, measuring hypervolume and Pareto spread.
- **Quantify Tier 2 selection accuracy** by measuring alignment between selected route characteristics and payload tag requirements across diverse mission types (medical sample, emergency medicine, disaster relief supplies).

- **Novelty:** The system uses **payload metadata as the sole input to Pareto point selection**, replacing entropy and diversity-based criteria that are blind to delivery context, with a deterministic graph-search Pareto front pre-computed once and reused across all missions.

Feasibility

- **Tools & Software Stack:** Python (NetworkX for graph-search, NumPy for objective computation, Matplotlib for Pareto front visualisation), ArduPilot / Mission Planner for UAV mission upload, MAVLink for ground-to-UAV communication, and a lightweight Tkinter or React dashboard for the operator
- **Hardware:** Quadcopter UAV with Pixhawk flight
- **Datasets:** Delivery region airspace graph constructed from OpenStreetMap and satellite imagery; obstacle and threat distributions simulated programmatically for controlled, reproducible benchmarking across all baseline comparisons.
- **Compute:** Tier 1 graph-search pre-computation runs entirely on a standard laptop with no GPU or cloud dependency; Tier 2 metadata-to-Pareto mapping is a near-instantaneous rule lookup with negligible compute cost.
- **Risks & Mitigations:** All algorithm validation and baseline benchmarking is completed through ArduPilot SITL (software-in-the-loop simulation) first, so real flight tests serve as confirmation of an already-validated system rather than the primary evaluation environment.

Project Timeline

Quarter	Key Activities	Deliverables
Q1 (Jul - Sep)	Literature review, problem formalisation, and airspace graph model design; Tier 1 multi-objective A*/Dijkstra implementation and unit tests	Literature review report, airspace graph model, Tier 1 A*/Dijkstra implementation with unit tests
Q2 (Oct - Nov)	Tier 1 Pareto front construction, objective normalisation, and front storage; Tier 2 payload metadata schema design and Pareto selection mapping	Pareto front construction module, objective normalisation scheme, payload metadata schema, Pareto selection mapping
Q3 (Dec - Jan)	Ground-station dashboard and mission file generator; system integration and ArduPilot SITL simulation validation; baseline comparison against A* and MOPSO with Pareto-front benchmarking	Ground-station dashboard, mission file generator, validated ArduPilot SITL integration, baseline and Pareto-front benchmarking results
Q4 (Feb - Mar)	Real outdoor flight tests and end-to-end performance measurement; results analysis, documentation, and final project report	Flight test data, end-to-end performance measurements, final thesis and project report